

SMART INDIA HACKATHON 2025

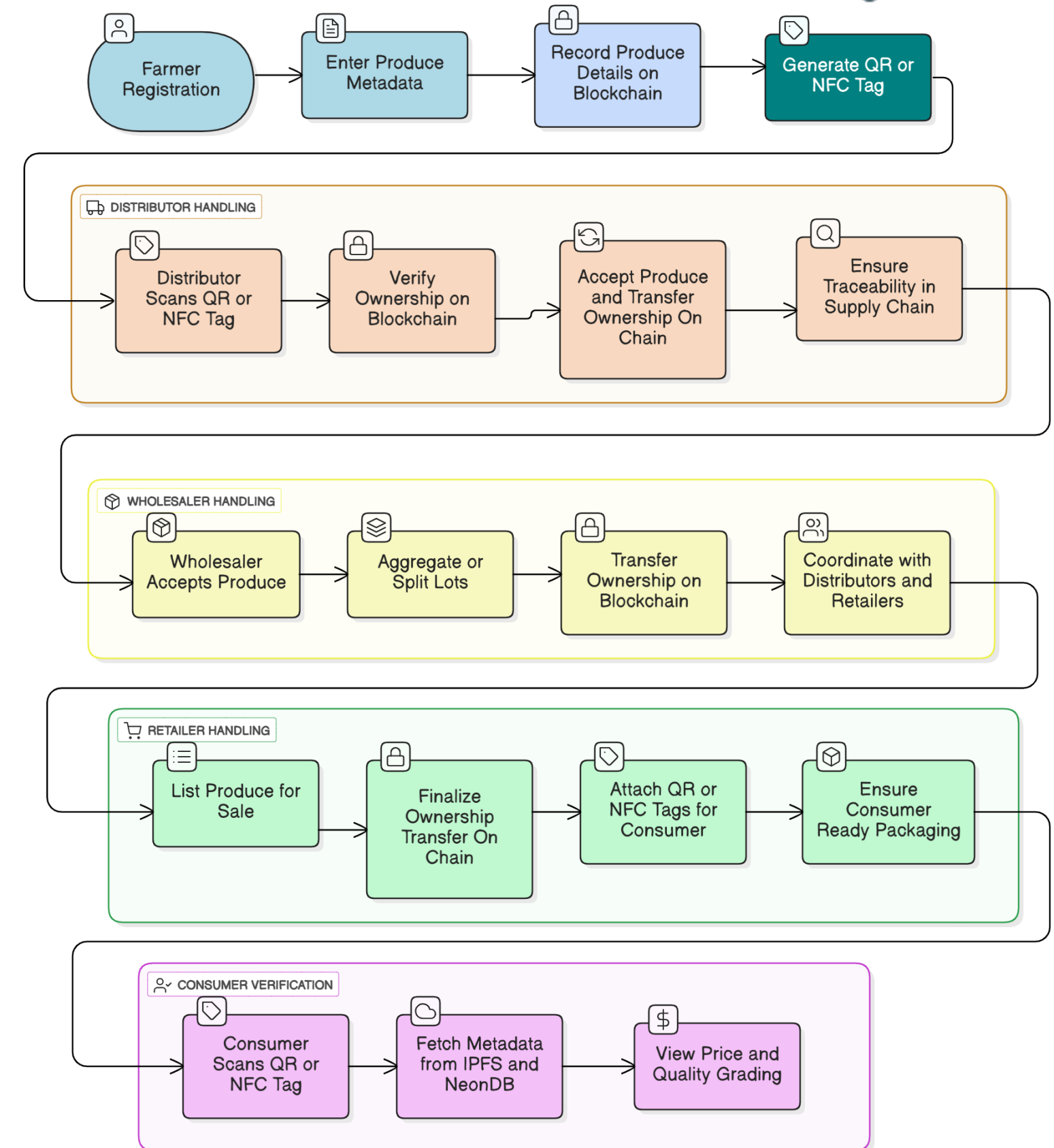


- Problem Statement ID - **25045**
- Problem Statement Title
**Blockchain-Based Supply Chain Transparency
for Agricultural Produce**
- Theme - **Agriculture and Food**
- PS Category- **Software**
- Team Name : **BitBytes**



Proposed Solution

Process Flow Diagram



Definition:

- Blockchain-powered web app with **role-based dashboards** (Farmer, Distributor, Wholesaler, Retailer).
- Integrated **ML microservices** for price prediction & produce grading.
- **QR / NFC verification** for consumers to check authenticity instantly.

Detailed Flow:

- Farmers **register produce** with metadata/images on IPFS (hash stored on-chain).
- **Ownership transfers** are logged **via smart contracts**, ensuring full traceability.
- **ML microservice** predicts **fair prices** and grades produce **quality**.
- Consumers scan **QR/NFC** to instantly verify origin, quality, and timeline.

How it Solves the Problem:

- **Transparent & Trustworthy** – Immutable records ensure fair pricing, quality, and provenance.
- **Fraud Prevention** – On-chain verification with off-chain metadata consistency.
- **Affordable & Scalable** – Low-cost Polygon deployment makes it viable for rural adoption.

Technologies Used:

- **Smart Contract** Deployed on Ethereum Testnets (Remix Solidity)
- **Frontend:** React with Tailwind CSS
- **Backend:** Node.js + Express.js
- **Database:** NeonDB (Postgres) with Prisma ORM
- **Integration:** APIs for mediator dashboards & real-time updates
- **Machine Learning:** Python FastAPI (RandomForest for price, MobileNetV2 for grading).

INFURA

SOLIDITY

node
JS

express

Python

GitHub

React

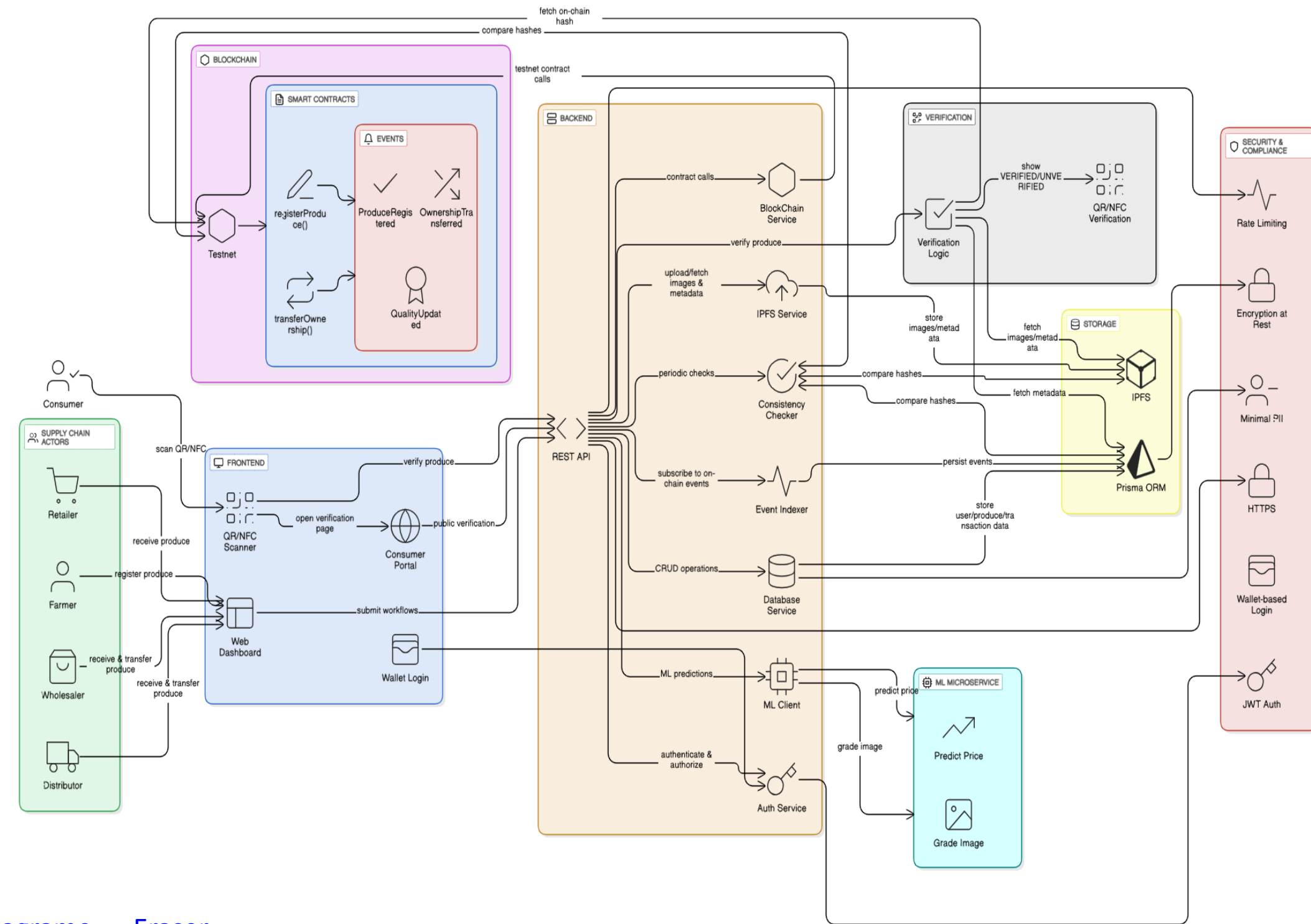
JS

React

ethereum

[Architecture Diagramme — Eraser](#)

System Architecture



Analysis of the Feasibility:

- **Scalable:** Dockerized for AWS/Heroku cloud deployment
- **Accessible:** Wallet login + mobile-friendly rural UI
- **Low-Cost:** Open-source stack reduces expenses
- **Interoperable:** API-ready for future agri-tech

Potential Challenges and Risks:

- Blockchain latency under high traffic
- Limited ML training data
- Low tech-literacy among farmers
- Data privacy & compliance concerns
- Wallet hacks / off-chain tampering

Strategies for Overcoming These Challenges:

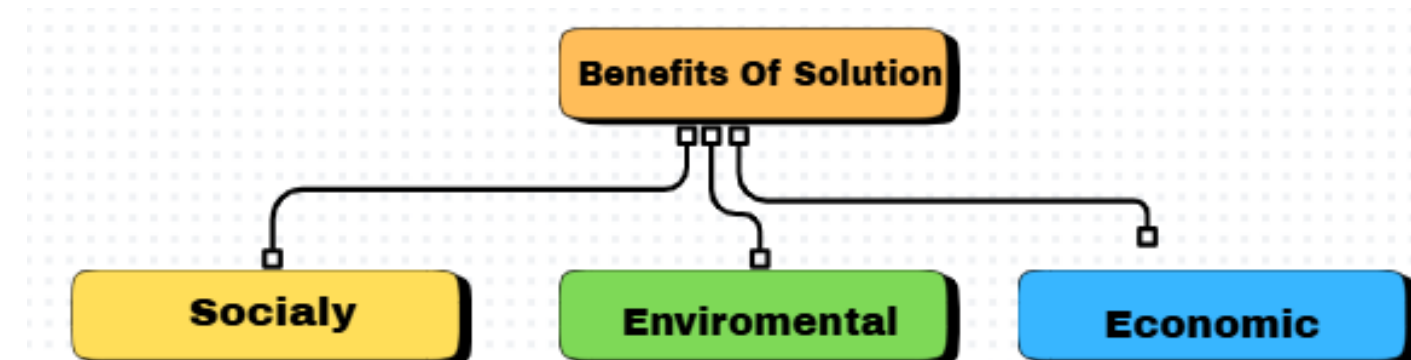
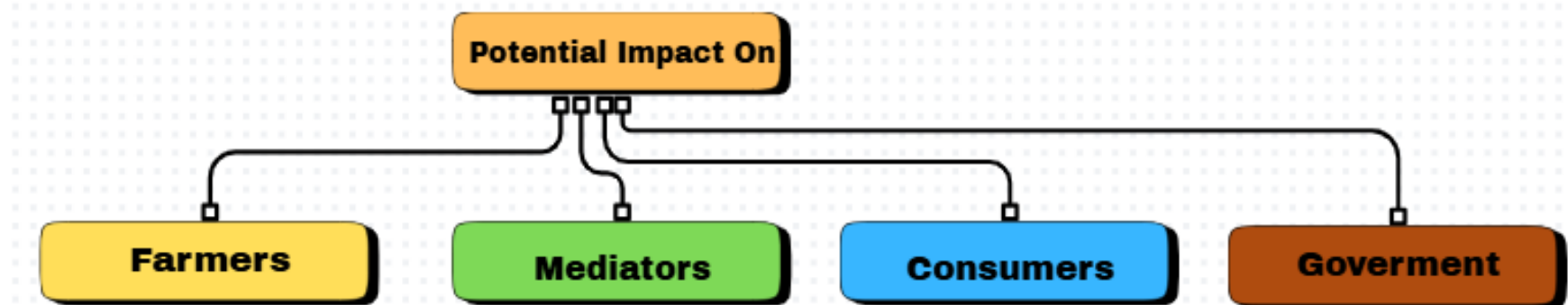
- Event indexing for faster blockchain queries
- Heuristic fallback + feedback-driven ML
- Simple manuals & guided onboarding
- JWT + HTTPS + regular consistency checks
- Rate limiting & reCAPTCHA for endpoints

Potential Impact on the Target Audience:

- **Farmers:** Fair pricing with ML predictions; traceable records for wider market access.
- **Mediators:** Streamlined transfers reduce disputes; real-time tracking improves inventory control.
- **Consumers & Government:** Verified origin and quality build trust; reliable data supports policymaking.

Benefits of the Solution:

- **Social:** Empowers small farmers, reduces food fraud, and ensures fair trade practices.
- **Economic:** Minimizes spoilage losses through grading; transparent pricing increases efficiency.
- **Environmental:** Monitors sustainable practices; reduces food waste with improved visibility.



Details/Links of the Reference and Research Work:

- **Blockchain in Agriculture:** “Blockchain for Agriculture” report by FAO.
- **ML for Pricing/Grading:** Kaggle agricultural datasets; MobileNetV2 research paper.
- **Storage & Verification:** IPFS integration with Pinata; ethers.js for QR/NFC.
- **Similar Implementations:** IBM Food Trust project; Odisha state agriculture policies.

Comparison with Existing Systems

Feature / Capability	AgriSupplyChain (Our Solution)	IBM Food Trust	VeChain	Provenance
End-to-End Role Coverage (Farmer → Distributor → Transporter → Retailer)	✓ (4 Dashboards)	✗ (Partial)	✗	✗
Transport Tracking (Dedicated Transporter Role)	✓	✗	✗	✗
Storage / Warehouse Tracking	✓	✗	✗	✗
QR/NFC Based Consumer Verification	✓	✓	✓	✓
ML-based Price Prediction	✓	✗	✗	✗
ML-based Image Grading (Quality Check)	✓	✗	✗	✗
Cost-Effective Infra (Polygon + IPFS + NeonDB)	✓	✗ (Proprietary + Expensive)	✗	✗
Public Blockchain Transparency	✓	✗ (Permissioned)	✓	✓
Role-based Dashboards	✓ (4 Dashboards)	✗	✗	✗
Open-Source Ready MVP	✓	✗	✗	✗